Suggested Answers for 2018 H2 Chemistry Preliminary Examination Paper 2

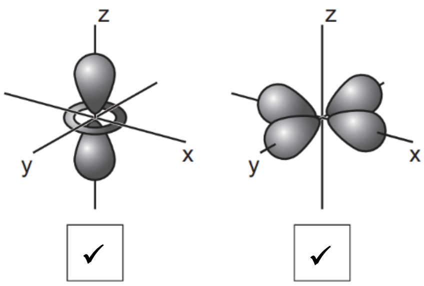
1 **(a)** % abundance of 5th isotope = 100 – (68.1 + 1.14 + 3.63 + 0.93) = 26.2 %

Let relative isotopic mass of 5th isotope be *x*.

$58.7 = 0.262x + (58 \times 0.681) + (61 \times 0.0114) + (62 \times 0.0363) + (64 \times 0.0093)$

$\therefore\ x = 60$

(b) **(i)**



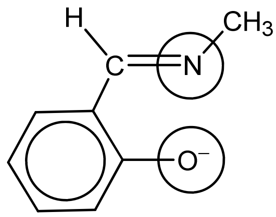
(ii) Visible light is absorbed when an electron transits from a lower energy d orbital to a higher energy d orbital that is partially filled.

Colour seen is the complement of the colours that are absorbed.

(iii) Colour: Green

Wavelengths corresponding to the blue and red regions are most absorbed (or green region is least absorbed).

(c)



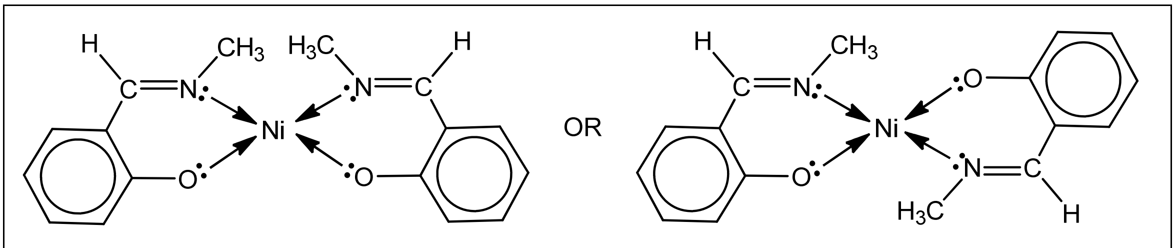
(d) $n(Ni^{2+}) = 4.0 \times 10^{-3} \times 3 \times 10^{-3} = 1.2 \times 10^{-5}$ mol

$n(X^-) = 6.0 \times 10^{-3} \times 4 \times 10^{-3} = 2.4 \times 10^{-5}$ mol

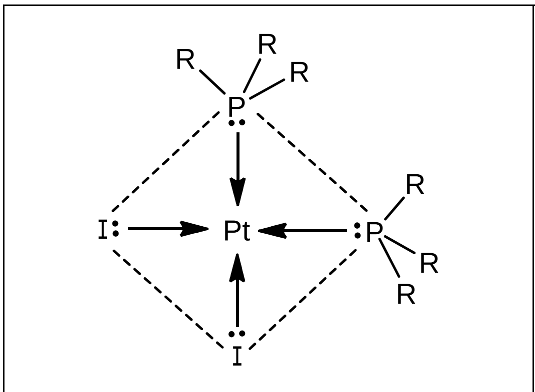
$n(Ni^{2+}) : n(X^-) = 1.2 \times 10^{-5} : 2.4 \times 10^{-5} = 1 : 2$

Empirical formula is NiX_2.

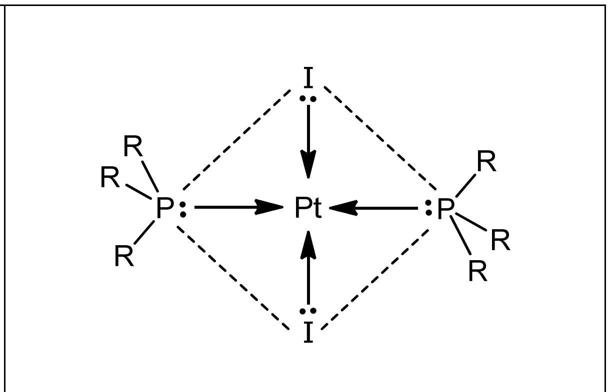
Structural formula of the complex:



(e)

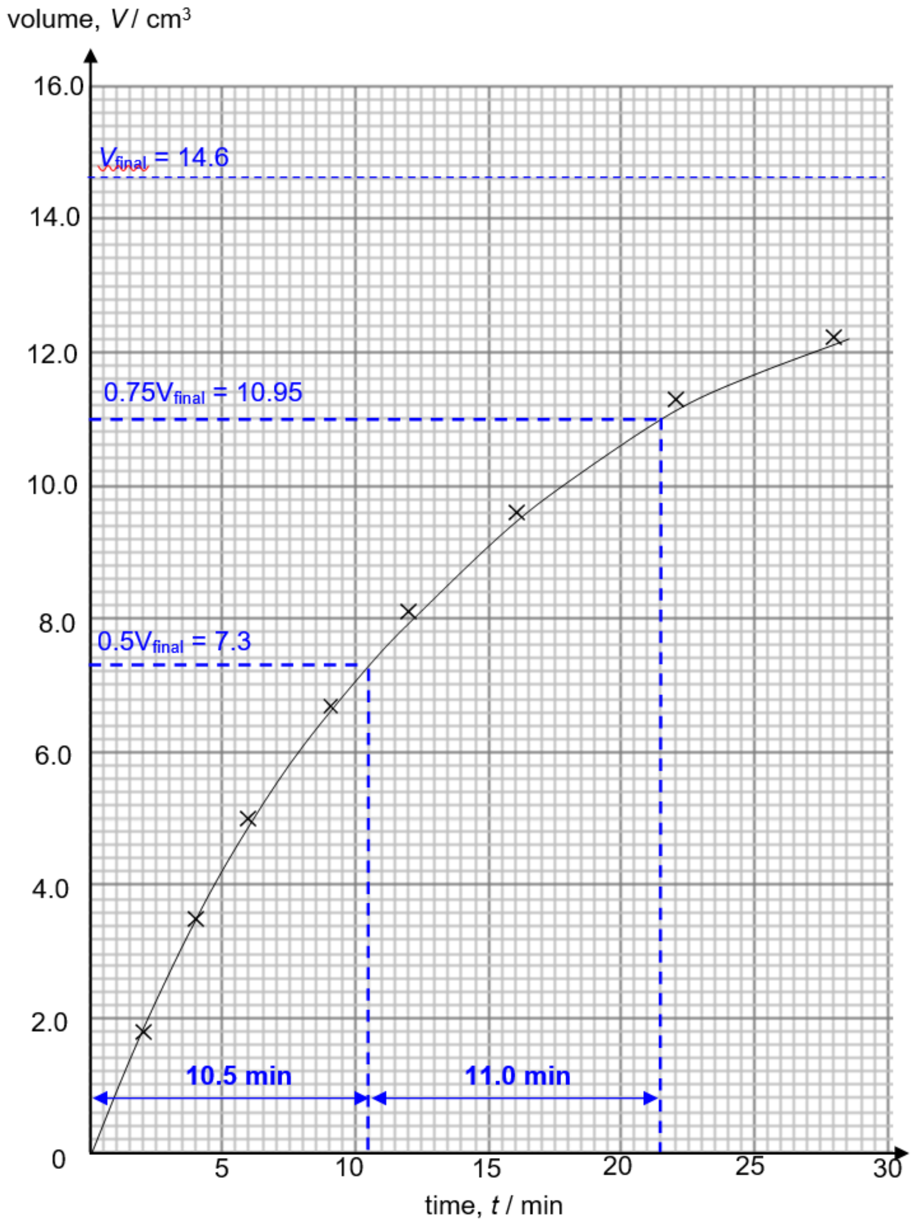


Isomer **A**



Isomer **B**

2 **(a)**



(i) Order of reaction with respect to benzenediazonium chloride: 1

1st $t_{½}$ = 10.5 min; 2nd $t_{½}$ = 11.0 min ⇒ Half-lives are relatively constant.

(ii) Average $t_{½}$ = 0.5 (10.5 + 11.0) = 10.75 min

For overall 1st order reaction, $k = \dfrac{\ln 2}{10.75}$ = 0.0645 min^{-1}

(iii) $n(N_2) = \dfrac{(101000)(14.7\times10^{-6})}{(8.31)(50+273)}$ = 0.000549 mol

= $n(C_6H_5N_2Cl)$ in 500 cm^3 solution

Original [benzenediazonium chloride] = $0.000549\times\dfrac{1000}{500}$ = 0.00110 mol dm^{-3}

(b) **(i)** type of reaction: electrophilic substitution

reagents: concentrated HNO_3

conditions: concentrated H_2SO_4, 50 °C

(ii) $1C_6H_5NO_2 + 7H^+ + 6e^- \rightarrow C_6H_5NH_3^+ + 2H_2O$

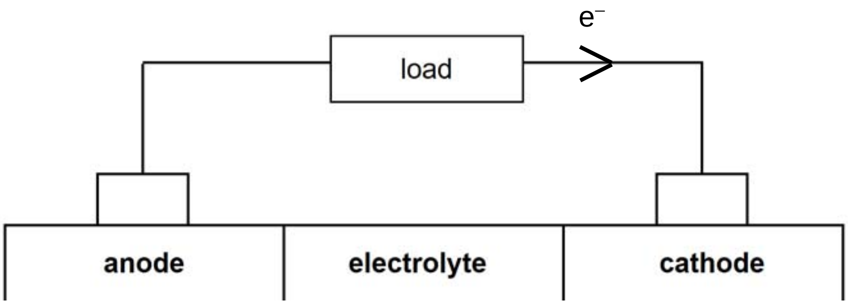
(iii) $2C_6H_5NO_2 + 3Sn + 14HCl \rightarrow 2C_6H_5NH_3Cl + 3SnCl_4 + 4H_2O$

(iv) NaOH will undergo acid-base reaction with $C_6H_5NH_3^+$ (or deprotonate $C_6H_5NH_3^+$) to give $C_6H_5NH_2$.

3 **(a)** Any 2 of the following:

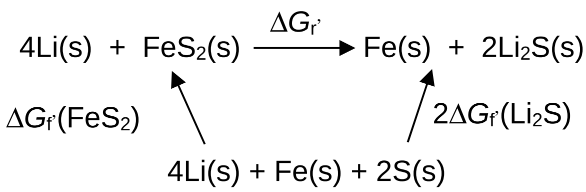
- Li is the lightest/has the lowest density so lithium battery is most portable.
- Li has the most negative $E^\ominus$/ is most easily oxidised among the metals and thus gives the largest e.m.f./$E^\ominus_{cell}$.
- Li has the largest electrochemical capacity so it can produce the largest amount of electrical charge per unit mass of the metal.

(b) **(i)**



(ii) Li is a reactive metal and will undergo redox reaction with water to produce H_2 gas which may cause explosion.

(c) **(i)**



$\Delta G_r^\ominus$ = Σ ($a_i \times \Delta G_f^\ominus$(products)) – Σ ($a_i \times \Delta G_f^\ominus$(reactants))

= 2(–439) – (–160)

= –718 kJ mol^{-1} (shown)

(ii) $\Delta G^\ominus = -nFE^\ominus_{cell}$

$-718 \times 10^3 = -(4)(96500)(E^\ominus_{cell})$

$E^\ominus_{cell} = +1.86$ V

(iii) $Li^+ + e^- \rightleftharpoons Li$ $\quad E^\ominus = -3.04$ V ---[O]

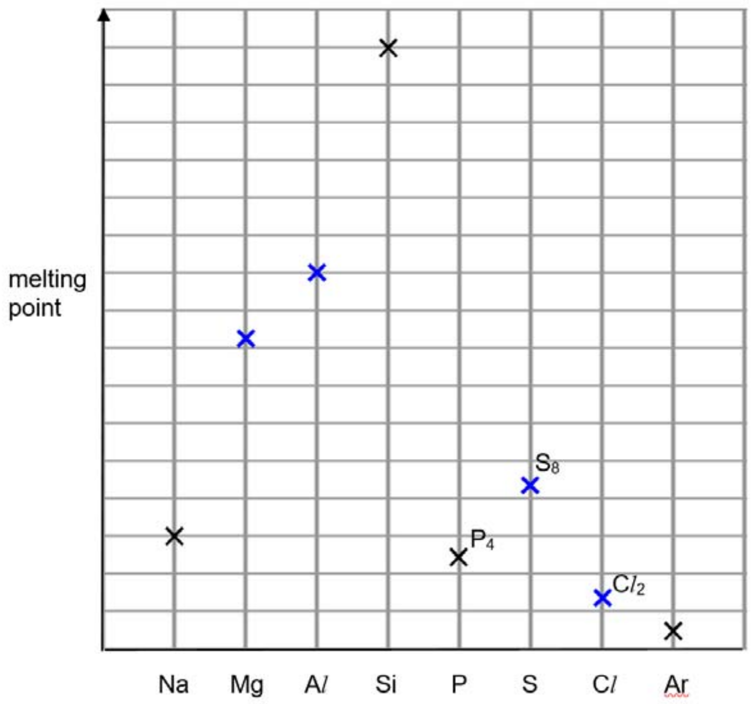
$+1.86 = E^\ominus(FeS_2/Fe) - (-3.04)$

$(FeS_2/Fe) = -1.18$ V

(iv)
- the left-hand electrode (anode): more positive
- The right-hand electrode (cathode): more positive

(v) $E^\ominus_{cell}$ will remain the same.

4 **(a)**



(b) **(i)** **D**: MgO **E**: $Mg(OH)_2$

(ii) X_2 is Cl_2.

$E^\ominus_{cell}$ for reaction of Cl_2 and Br^- = (+1.36) – (+1.07) = +0.29 V > 0 (energetically feasible).

OR

$E^\ominus_{cell}$ for reaction of I_2 and Br^- = (+0.54) – (+1.07) = –0.53 V < 0 (energetically not feasible).

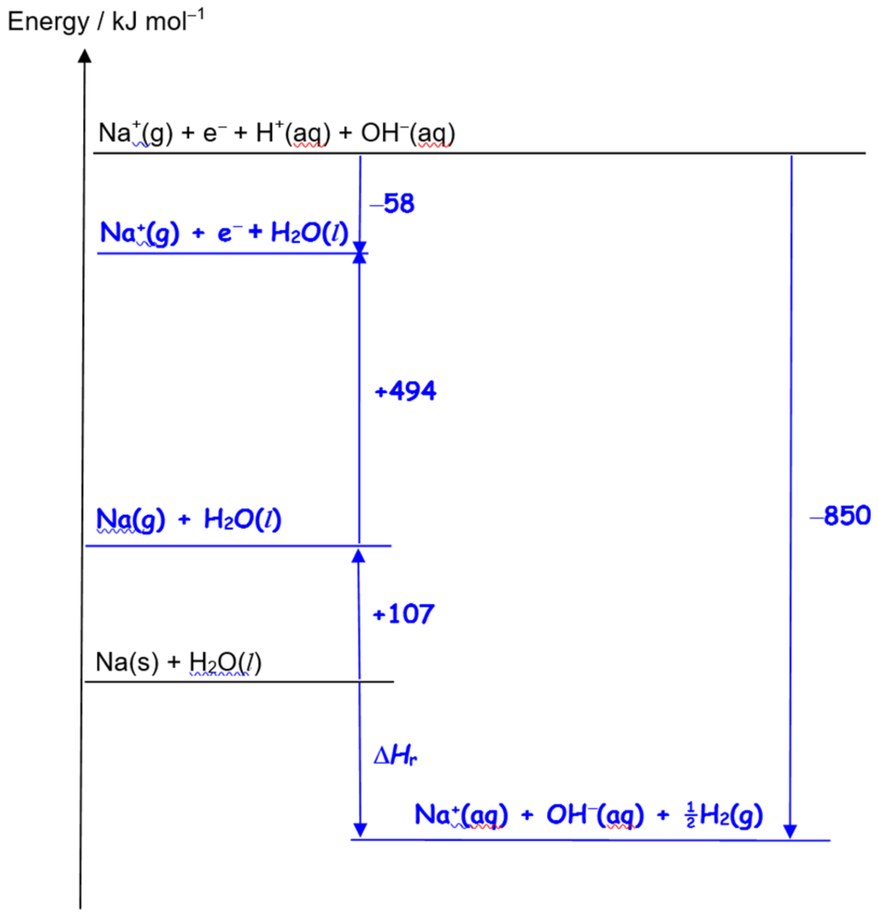
OR

$E^\ominus(Cl_2/Cl^-) > E^\ominus(Br_2/Br^-)$ so Cl_2 is a stronger oxidising agent than Br_2 and thus can oxidise Br^- to Br_2.

OR

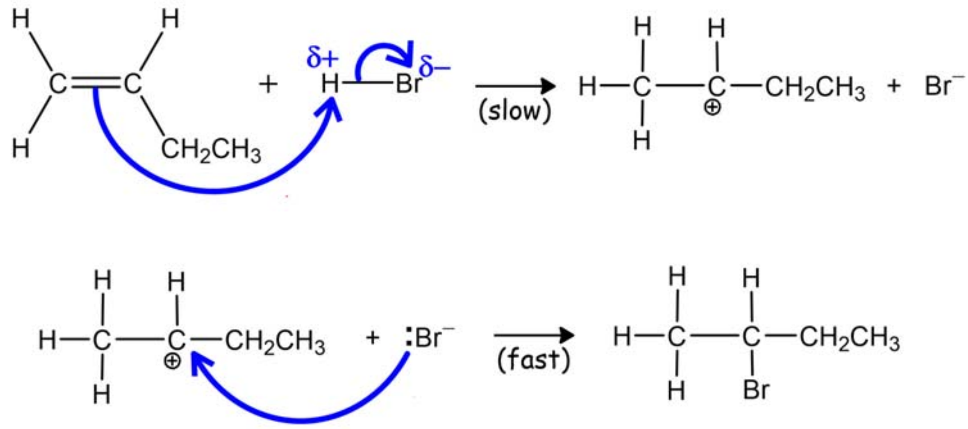
$E^\ominus(Br_2/Br^-) > E^\ominus(I_2/I^-)$ so I_2 is a weaker oxidising agent than Br_2 and thus cannot oxidise Br^- to Br_2.

(c)



ΔH_r = (+107) + (+494) – (–58) + (–850) = –191 kJ mol^{-1}

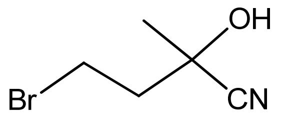
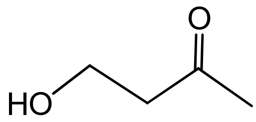
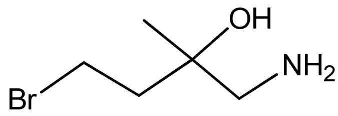
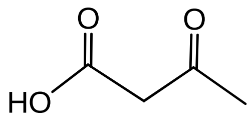
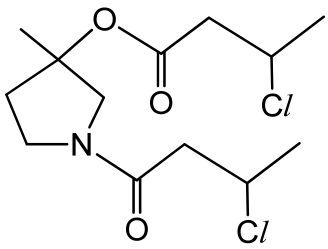
5 (a) (i) Name of mechanism: electrophilic addition



(ii) In the first step, secondary carbocation, $^{+}CH(CH_3)CH_2CH_3$, (which yields 2-bromobutane) is more stable and thus more readily formed than primary carbocation, $^{+}CH_2CH_2CH_2CH_3$, (which yields 1-bromobutane) as it has one more electron-donating alkyl group which disperses its positive charge more.

(iii) There is equal probability for Br^- to attack either side of the trigonal planar C^+ of carbocation in step 2, forming a racemic mixture.

(b) **(i)**

J: 	**L:** 
K: 	**M**: 
N: 	

(ii) reaction 1: HCN, trace amount of KCN/NaCN

reaction 2: $LiAlH_4$, dry ether OR H_2, Ni, heat

reaction 4: aqueous NaOH/KOH, heat

reaction 5: acidified $K_2Cr_2O_7$/$KMnO_4$, heat

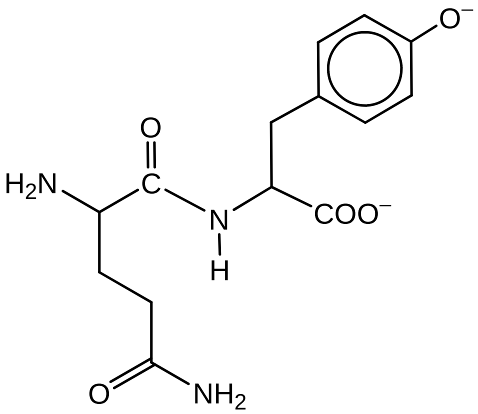
reaction 6: $NaBH_4$, methanol OR H_2, Ni, heat

(iii) reaction 7: nucleophilic substitution

reaction 8: condensation

6 **(a)** $M_r = 146 + 155 + 131 + 105 + 181 - (4 \times 18.0) = 646$

(b) **(i)**



(ii) spot **P**: tyr

Tyrosine has a charge of 2– at pH = 12 and a smaller mass/M_r than gln-tyr. It has the highest $\frac{\text{charge}}{\text{mass}}$ ratio compared to the other two species so it moves the fastest and thus furthest from original position.

(iii) Charge of gln-tyr is twice that of gln and its mass/Mr of is about twice that of gln.

OR

gln and gln-tyr have similar $\frac{\text{charge}}{\text{mass}}$ ratios ($\frac{\text{charge}}{\text{mass}}$ of gln is $\frac{1}{146} = 0.00685$ whereas $\frac{\text{charge}}{\text{mass}}$ of gln-tyr is $\frac{2}{309} = 0.00647$).

(c) test: Br_2(aq)

observations: Tyr decolourises orange Br_2(aq) and forms a white precipitate while gln will not.

OR

test: neutral $FeCl_3$(aq)

observations: Tyr will give a violet colouration while glu will not.

OR

test: NaOH(aq), heat

observations: Gln will give NH_3 gas which turns damp red litmus paper blue while tyr will not.

(d) **(i)** sp^2

(ii) N_b can act as a Bronsted base.

This is because the lone pair of electron on N_a is in an unhybridised p orbital which is parallel to the adjacent π electron systems so it is delocalised into C=C and C=N_b due to p-p orbital overlap. Hence, the lone pair of electrons is not available for protonation.

However, the lone pair (of electron) on N_b is in a sp^2 orbital which is on the same plane as the ring (or not parallel to adjacent π electron system) and hence it will not be delocalised into the adjacent C=C. Thus, the lone pair of electron is available for protonation.